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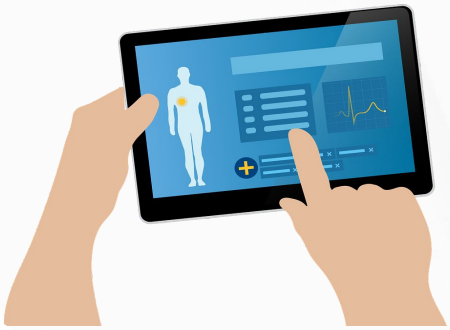
# COMPUTER-IMPLEMENTED METHOD FOR SECURE MANAGEMENT OF MEDICAL RECORDS IN AN AI-ENHANCED SYSTEM



## ✓ KEY INFORMATION

**Patent Number: PI 2025007955**

**Technology Readiness Level: TRL 5**

**Market Readiness Assessment: MR 5**

**Location: Universiti Malaya, Malaysia**

## ✓ TECHNOLOGY OVERVIEW

This technology is a **secure, AI-enhanced digital medical records management system** designed to help healthcare providers, platforms, and digital health companies **store, share, and use medical records safely and efficiently**.

The system combines **strong data security**, **controlled access management**, and **artificial intelligence** to support modern healthcare workflows. It allows medical records to be uploaded, securely stored, shared on a time-limited basis, and intelligently summarized, improving accessibility while maintaining strict data protection standards.

The solution is designed for **cloud-based deployment** and integration into existing healthcare or health-tech platforms.

## ✓ FEATURES AND SPECIFICATIONS

- Encrypted cloud storage for secure medical record management
- Mobile and web-based interfaces for record upload and access
- Optical character recognition (OCR) for digitising scanned medical documents
- Role-based access control to manage viewing, sharing, and usage permissions
- Time-limited, permission-controlled record sharing via secure access links
- Secure viewer with view-only restrictions to prevent unauthorised copying
- AI-powered record summarisation for faster clinical or administrative review

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- Intelligent chatbot interface for contextual, query-based record navigation
- Emergency access mode to allow temporary authorised access when needed

## ✓ POTENTIAL APPLICATION

- Hospitals, clinics, and specialist medical centres
- Telemedicine and digital healthcare platforms
- Health information exchanges and medical data hubs
- Insurance and claims processing systems
- Corporate healthcare and occupational health providers
- Health-tech startups and SaaS healthcare solutions
- Cross-border healthcare data sharing (subject to regulations)

## ✓ UNIQUE VALUE PROPOSITIONS

- Balances **strong data security** with **ease of access** for authorised users
- Reduces administrative workload through **AI-generated summaries**
- Improves collaboration with **safe, time-limited record sharing**
- Enhances compliance with healthcare data protection standards
- Scalable, cloud-ready design suitable for enterprise deployment
- Enables faster decision-making without exposing sensitive data
- Attractive for **licensing, white-label solutions, and platform integration**

**YOU HAVE INQUIRY?**

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